



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS -
JAN/FEB 2023

SUBJECT : CSE 2153 - DIGITAL SYSTEM DESIGN
(MAKEUP)

DIGITAL SYSTEM DESIGN [CSE 2153]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Reduce the number of states in the following state table and tabulate the reduced state table. Compare the output sequence for the given table and reduced state table using the input sequence starting from state a, and input sequence 110010011.

Table 1: State table

Present State	Next State		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

(5)

- 1B) Simplify the following function as sum of products, using K-map technique $f(A, B, C, D)$
 $= \sum m(0, 1, 4, 8, 9, 10)$ (3)

- 1C) Explain the principal components of an ASM Chart with its symbols. (2)

- 2A) It is required to design a digital system with the operations shown in Table P.2A (5)

Table P.2A

Operation	Function Performed
I0: Load R0, Data1	$R0 \leftarrow \text{Data1}$
I1: Mul R0, Data2	$R0 \leftarrow [R0] * \text{Data2}$

There are four registers R0 to R3 in the system which are connected to the bus through tri state drivers. External data can be loaded into the registers from the Data input, which is connected to the bus using tri state drivers enabled by the Extern control signal. The system also includes a multiplier unit. One of its data inputs is provided by a register A, that is attached to the bus, while the other data input B, is directly connected to the bus. It also has a temporary output register G. The operation is initiated by setting an input w to 1, and the control circuit asserts the *Done* output when the operation is completed.

- Draw the block diagram for this system.
- Determine the timing signals required.
- Design the control circuit.

- iv) Tabulate the necessary control signals required in different timings.
 v) Derive the expressions for the outputs of the control circuit in terms of operations and timing signals.

2B) Construct 3 to 8 decoder using 2 to 4 decoders and other necessary gates. Write the Verilog code for the 2 to 4 decoder using case statement. Using this module write the Verilog code for the above design of 3 to 8 decoder. (3)

2C) Analyze the state diagram shown in Fig P.2C and obtain the excitation table using SR flip flops. (2)

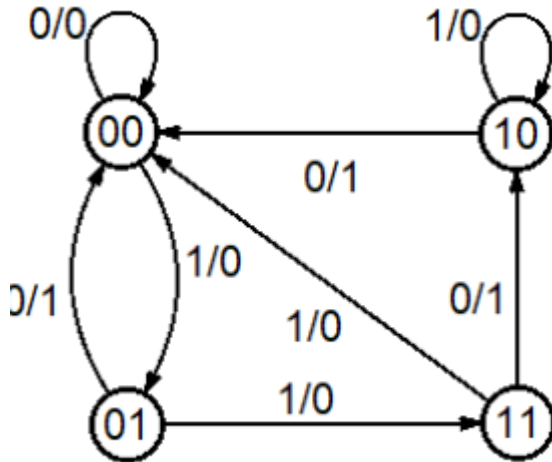


Fig P.2C

3A) Construct 4-bit synchronous binary up-counter using JK flip flops. Draw the waveform for the same. (5)

3B) Realize a JK flip-flop using AND and NOR gates. (3)

3C) Demonstrate the working of tri-state drivers. (2)

4A) Build the algorithm for bit counting circuit and draw its ASM chart. With respect to this, differentiate traditional flow chart and ASM chart. Also design the datapath circuit for the same and draw the ASM chart for its control circuitry. (5)

4B) Design a 48-bit carry lookahead adder with ripple carry between 8-bit carry lookahead blocks. Determine the gate delay involved to generate the final carry. (3)

4C) Design a 5-bit shift register with parallel load that operates according to the function table shown in Table P.4C.

Table P.4C

S1	S0	Register Operation
0	0	Load Parallel data
0	1	Rotate right
1	X	Shift left

5A) Design and explain the clocked NAND gate based RS flip flop with necessary diagram. Derive its characteristic equation. Illustrate the method to overcome the intermediate state in the RS flip flop by adding only an additional gate. (5)

5B) Construct NAND gate using CMOS circuit and explain its operation. (3)

5C) Develop an asynchronous sequential circuit to generate 100 KHz and 50 KHz signal when 200 KHz clock signal is given using T flip flops. Draw the timing diagram of all (2)

the three signals.

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